Techniques and algorithms checklist

Every implemented method, where it lives, why it is there, the formula, and what it must produce. No week labels — this is the software as built.

Read each row as a contract: **where** (module and file), **step and purpose**, **reason**, **formulae**, **expected output**. Module 2 rows 11–27 are the S21 teaching track plus the general MATLAB track and the Module 3 handoff. Module 3 and ROI start after that. Output is not a clinical diagnosis.

Module 1 — Data acquisition

Module / where	Step and purpose	Reason	Formulae and definitions	Expected output
M1 · loader.py	1. Dispatch the file by extension	One entry point for the GUI and automation. Later modules never need to know the original format.	ext = lower(file suffix). .mat → MATLAB/BMID loader. .s1p/.s2p/.s4p/.s8p → Touchstone loader. Else UnsupportedFileFormatError.	Call to the correct parser.
M1 · validator.py	2. Check the path before parsing	Avoid crashing on a missing, empty or unknown file.	File must exist, size > 0, extension ∈ {.mat,.s1p,.s2p,.s4p,.s8p}.	OK to parse, or a typed error.
M1 · touchstone_loader.py	3. Import and inspect the Touchstone header; extract S21 on .s2p	Identify the frequency unit, data format and reference impedance. Imaging uses transmission S21, not the whole 2×2, on the UG track.	f_Hz = f_GHz×10■ ; f_Hz = f_MHz×10■ ; f_Hz = f_kHz×10³. Option line: # UNIT S {RI|MA|DB} R Z0. Row order after f: S11, S21, S12, S22.	Header (unit, RI/MA/DB, Z0) and extracted S21 plus the full (n_f, n_ports, n_ports) matrix.
M1 · touchstone_loader.py	4. Convert S21 to complex linear form	Preserve magnitude and phase. Do not average, filter or subtract dB values.	DB: A=10^(D/20), ϕ=ϕ_deg·π/180, S21=A(cosϕ+j sinϕ). MA: S21=mag(cosϕ+j sinϕ). RI: S21=R+jI. j²=−1.	Raw complex S21 in linear units.
M1 · matlab_loader.py	5. Load legacy or v7.3 MATLAB files	Lab benches export both old MAT and HDF5 v7.3. Complex v7.3 arrays arrive as real/imag fields.	If h5py.is_hdf5: visit datasets and convert {real,imag} → real+j imag. Else scipy.io.loadmat. Drop keys starting with __.	Dictionary of numeric arrays.
M1 · matlab_loader.py	6. Find the frequency vector and S-parameter array	MATLAB files have no single schema.	Name hints: freq/frequency/… ; else longest real increasing vector. If max(f)<1000 treat as GHz: f_Hz=f×10■. S: name hints s_param/s21/… plus dim matching n_f. Known cube fallback: fd_data_s21 → 1–8 GHz, 1001 points.	f in Hz and S oriented with frequency on axis 0.
M1 · bmid_loader.py	7. Slice one UM-BMID scan from the 3-D cube	A cube is many experiments. Loading without an index would mix them.	cube[scan, freq, ant] ∈ ■. S = cube[k,:,:], shape (1001, n_ant). f=linspace(1e9,8e9,1001). Pair md_list_* → labels tum_diam/tum_rad, tum_x, tum_y, birads, ant_rad.	One MicrowaveDataset plus tumor/healthy metadata in SI units.
M1 · physical_metadata.py	8. Parse radius, wave speed and FOV from comments	Reconstruction needs geometry even when it is only written in a note.	Regex for antenna radius (m/cm), wave speed (m/s), x span / y span. cm → m by /100.	metadata: antenna_radius_m, wave_speed_m_per_s, reconstruction_*_span_m.
M1 · validator.py · dataset_info.py	9. Validate frequencies and S; build MicrowaveDataset	Finite, non-empty arrays only. Downstream code uses one object for every format.	f finite, S finite, no NaN/Inf. n_samples = |S|. Summary: n_freq, range, n_ports, size, variable names.	MicrowaveDataset + DatasetSummary for the GUI panel.

Module 2 — Signal preprocessing and handoff

Module / where	Step and purpose	Reason	Formulae and definitions	Expected output
M2 · preprocessing_pipeline.py	10. Route .s2p versus general / BMID	The UG brief is S21-only. BMID _adi is already cleaned and must not be cleaned again.	If Touchstone and .s2p → S21 track. Else five-stage general loop. BMID preset: every method = none.	Chosen track and a PreprocessingConfig.
M2 · touchstone_s21.py	11. Validate and sort samples	Prevent invalid values, duplicates and reverse frequency order from breaking IFFT.	N_f = N_S21. Sort so f[k+1]>f[k]. Drop/repair non-finite samples. Record notes.	Ascending frequency and S21 vectors of equal length.
M2 · touchstone_s21.py	12. Check uniform spacing; interpolate only if required	A conventional IFFT needs constant Δf. Matched target/reference must share one grid.	Δf[k]=f[k+1]−f[k]. If not uniform: Δf_u=(f_max−f_min)/(N_u−1), f_u[k]=f_min+k Δf_u, R_u=Interp(f,R,f_u), I_u=Interp(f,I,f_u), S_u=R_u+j I_u. Linear interpolation on Re and Im.	Uniform f and complex S21. Flag interpolation_performed.
M2 · touchstone_s21.py	13. Build magnitude and phase representations	Inspect the original signal. Keep both wrapped and continuous phase.	S_dB=20 log10 |S|. ϕ_w=atan2(I,R). ϕ_u=unwrap(ϕ_w). Unwrap fixes ±π jumps; it does not denoise.	Magnitude dB, Re, Im, wrapped phase, unwrapped phase (GUI plots).
M2 · touchstone_s21.py	14. Detect isolated invalid samples (Hampel / median / local)	Stop one glitch from becoming a broadband artefact after IFFT.	m_k=median{x[r]: r∈W_k}. MAD_k=median{|x[r]−m_k|}. σ■_k=1.4826 MAD_k. Flag if |x[k]−m_k|>λ σ■_k (λ commonly 3). Replace flagged Re/Im. Default method: hampel.	Corrected complex S21 and a count of repaired samples.
M2 · touchstone_s21.py	15. Average repeated sweeps when available	Reduce uncorrelated noise. Do not average different antenna positions or phantoms.	S■_21[k]=(1/M) Σ_m S_21^(m)[k]. M = number of identical-condition repeats.	Complex-averaged S21, or skip with Averaging=No.
M2 · touchstone_s21.py	16. Optional reference subtraction in frequency	Remove a matched empty / tumor-free sweep when the user supplies one.	S_diff[k]=S_T[k]−S_R[k]. Same grid, same preprocessing.	Reference-subtracted S21, or skip.
M2 · filtering.py · touchstone_s21.py	17. Apply a mild noise filter on Re and Im	Reduce point-to-point noise without wiping a weak target bump. Never filter dB as if it were linear.	Default .s2p: Savitzky–Golay, window L=7, polyorder=3, on R and I. Also available: moving average R_f[k]=(1/L)Σ R[k+r], Butterworth, median, Gaussian. S_f=R_f+j I_f.	Filtered complex S21. GUI comparison plots.
M2 · touchstone_s21.py	18. Quantify how much filtering moved the signal	Check that cleaning did not rewrite the measurement.	NRMSE = ‖S_f − S_before‖_2 / ‖S_before‖_2. Measures alteration, not image quality.	NRMSE in the validation report; chosen filter name and parameters.
M2 · week3_time_domain.py	19. Hamming window (keep unwindowed copy)	Taper band edges so IFFT sidelobes / ringing drop. Reconstruction still defaults to filtered S21(f).	w_H[k]=0.54−0.46 cos(2πk/(N−1)), k=0…N−1. S_H[k]=S_f[k] w_H[k].	S21_filtered and S21_Hamming both stored.
M2 · week3_time_domain.py	20. Window target and reference identically	Subtraction is only valid if both sides saw the same window.	S_{T,H}=S_{T,f} w_H and S_{R,H}=S_{R,f} w_H. Same w_H, same N.	Matched windowed pair, or note that no reference was given.
M2 · ifft.py · week3_time_domain.py	21. IFFT to a complex time response	Delay-domain signals for clutter work and for DAS/DMAS.	s[n]=(1/N) Σ_k S[k] e^{j 2π kn/N} (NumPy ifft). Δt=1/(N Δf), t[n]=n Δt, T_period=1/Δf, B=f_max−f_min=(N−1)Δf. Need uniform Δf (rtol 1e-3).	time_s and time-domain traces with and without Hamming.
M2 · week3_time_domain.py	22. Matched reference subtraction in time	Suppress the tumor-free / system response. IFFT is linear so subtract-in-f or subtract-in-t is the same if windows match.	s_diff[n]=s_T[n]−s_R[n] and IFFT(S_T−S_R)=IFFT(S_T)−IFFT(S_R).	Reference-subtracted time (and frequency) copies when a reference exists.
M2 · week3_time_domain.py	23. Group-mean clutter removal, only if ≥2 channels	Subtract the response common to a set of comparable antennas. A single .s2p has P=1 so this is skipped.	μ_G[n]=(1/P) Σ_p s_p[n], s_clean[n]=s_p[n]−μ_G[n].	Clutter-reduced channels, or a skip note.
M2 · week3_time_domain.py	24. Optional global normalize	One scale for all channels. Do not normalize each channel on its own (that would destroy relative amplitudes).	α = max_{p,n} |s_clean|, s■ = s_clean / α.	Original-scale and globally normalized copies; α stored.
M2 · preprocessing_pipeline.py	25. General-track filter → calibrate → normalize → background → artifacts	Used for multi-trace MATLAB that is not a BMID pass-through.	Calibration: S_cal=S_meas/S_ref, or self-cal: subtract per-frequency mean across traces. Normalization (phase kept): max |S|, min–max |S|, or z-score |S|. Background: S−S_bg. Artifacts: hybrid / other methods in artifact_suppression.py.	Processed S of the same shape; stage_outputs snapshots.

Module / where	Step and purpose	Reason	Formulae and definitions	Expected output
M2 · preprocessing_pipeline.py	26. Signal-quality report before vs after	Tell the GUI whether cleaning helped.	SNR_dB=10 log10(P_smooth / P_residual) with 5-sample moving-average “signal”. DR_dB=20 log10(max|S|/min|S|). Artifact reduction = 1 − E_after/E_before.	SignalQualityReport numbers on the Processing Summary panel.
M2 · handover_export.py	27. Validate and export the Module 3 handoff	Give reconstruction everything it needs, with traceability.	Check equal lengths, uniform f, finite complex values, matching metadata. Write .mat / optional .csv / .json.	frequency_Hz, time_s, S21_raw, S21_filtered, S21_Hamming, S21_reference_subtracted, S21_clutter_removed (if any), Tx/Rx coordinates, labels, processing_parameters, validation report.

Module 3 — Reconstruction

Module / where	Step and purpose	Reason	Formulae and definitions	Expected output
M3 · reconstruction_manager.py	28. Infer geometry (radius, c, FOV, array)	Pixels and delays are meaningless without a physical layout.	x=linspace(x_min,x_max,n_x), y=linspace(y_min,y_max,n_y), default 64×64. a_i=(r cos θ_i, r sin θ_i). 360°: endpoint=False. 355° BMID arc: endpoint=True. Defaults: r=0.08 m, c=3e8, FOV ±5 cm. BMID: r=0.18 m, FOV ±6 cm. Optional r_eff=0.97(r–0.106)+0.148 m.	ReconstructionConfig and antenna positions (N×2 m).
M3 · ifft.py	29. Frequency → time for every antenna trace	Beamformers sample delays, not frequencies.	time_signals = ifft(S, n=n_f+pad, axis=0). t[n]=n/(N Δf). Fail if Δf is not uniform.	Complex (N_time, n_traces) array.
M3 · das.py	30. Delay-and-Sum (DAS)	Add each antenna's envelope at the two-way delay of that pixel. Coherence weight fights ring clutter.	d_i=‖a_i−r‖, τ_i=2 d_i/c, n=round(τ_i/Δt). I_plain=Σ |s_i(τ_i)|. CF=|Σ s_i|² / (N Σ |s_i|²). I_DAS=I_plain · CF^γ, γ=0.55 (γ=0 is textbook DAS; cohort average).	One real image (n_y, n_x).
M3 · dmas.py	31. Delay-Multiply-and-Sum (DMAS)	Keep energy that is consistent on two antennas at once. Sharper than DAS, slower.	p_ij=s_i conj(s_j). signed_power(z,p)=Re{sign(z)(|z|+ε)^p}, p=0.30, ε=1e-9. I_DMAS=Σ_{i<j} signed_power(p_ij). Cohort average; textbook p=0.50.	One real image.
M3 · dmas_d4.py	32. DMAS-D4 contrast map	Compress dynamic range after DMAS. Cohort sweep keeps the textbook exponent 0.25. 0.55 was not better on average.	I_D4 = sign(I_DMAS) · |I_DMAS|^{0.25}.	One real image.
M3 · quality/metrics.py	33. Image quality metrics	Compare the three pictures with numbers, not only by eye.	SNR=mean(I)/std(I). SCR=max(I)/mean(|I−mean(I)|). CCR=(mean(top 5%)−mean(rest))/std(rest). contrast=(max−min)/(max+min). FWHM_px = mean of half-max widths on the peak row and column.	A metrics dict per image.
M3 · beamformer_selector.py	34. Select which beamformer to keep	One image goes to ROI. Mode depends on labels and the user.	Quality: score=0.35 SNR_n+0.35 SCR_n+0.20 contrast_n+0.10(1−time_n), metrics min–max normalized across the three. Also: prefer DMAS-D4, force DAS/DMAS/D4, or tumor_gt (closest ROI to labeled (x,y)).	Selected name + score table + reason string.
M3 · auto_calibrate.py	35. Auto Tweak: search geometry + beamformer	Wrong angle/flip/arc looks like a missed tumor. Search is only applied if it clearly beats the current settings.	Sweep angles {0,90,180,270}, flips, CW/CCW, 360/355°, phase-delay on/off, ROI modes. DAS first, then top-8 full beamformers. GT score=1/(dist+1e-4). Apply if GT error drops by ≥0.5 mm.	Best config, or keep the baseline.

ROI, quality labels, and automation

Module / where	Step and purpose	Reason	Formulae and definitions	Expected output
ROI · roi_detector.py	36. Normalize the selected image	Threshold 0.70 must mean the same thing on dim and bright images.	I_n=(I−I_min)/(I_max−I_min+ε), ε=1e-12.	I_n ≈ [0,1].
ROI · roi_detector.py	37. Gaussian smooth	Stop a 1-pixel spike becoming its own region.	I_s=Gaussian(I_n, σ=1.0 px).	Smoothed image.
ROI · roi_detector.py	38. Threshold + morphological opening	Turn intensity into candidate pixels, then drop speckle.	B=1{I_s≥T}, T=0.70. Opening with a 3×3 ones kernel.	Binary map B.
ROI · roi_detector.py	39. Connected components	Separate islands are separate candidates.	Label connected 1s. Drop area < 6 px. If none: raise T by 0.05 (max 0.95) or box the global peak.	List of components / bounding boxes.
ROI · roi_detector.py	40. Score each component and pick spots	Prefer a compact interior hotspot, not the largest splash or an edge ring.	Score=(0.65 I_peak+0.35 I_mean)×(A/A_bbox)×1/(1+10 AreaRatio)×γ_edge, γ_edge=0.65 if the box touches the border else 1. Off-centre extra factor (0.2+0.8 r_norm)(1+1.5 r_norm). Keep ≤4 spots, gap ≥6 px, score ≥0.40×best (auto 0.28). GT may add the nearest peak within 3 cm.	Ranked ROI list; primary = highest score.
ROI · reconstruction_manager.py	41. High-resolution rebuild inside the box	Same physics, finer pixels, smaller FOV.	Convert box to metres. n=clip(box×4, min 96, max 256). Re-run DAS or DMAS or D4 on that span only.	ROIRefinement image + local spans.

Module / where	Step and purpose	Reason	Formulae and definitions	Expected output
ROI · characterization.py	42. Characterize the selected region	Describe size and focus in centimetres.	centroid via FOV interpolation. d_eq=2√(A_cm²/π). local SCR on the box. FWHM_cm=FWHM_px×px_cm. compactness=A_px/(w h). eccentricity=(aspect−1)/(aspect+1).	TumorCharacterization numbers.
ROI · roi_detector.py	43. Tumor-candidate suspicion and YES/NO	Winning the ROI contest is not the same as looking tumor-like. Not a diagnosis.	local_contrast=peak/mean_outside. ring_excess=peak/median(same radius). Suspicion=clip(0.38 tanh((lc−1)/2)+0.22/(1+(AR/0.08)²)+0.15(0.45+0.55 r_norm)+0.25 tanh((ring−1.15)/0.35),0,1). Conf=clip(Susp×0.85_edge×0.65_big×0.55_lowring,0,1). YES if Conf≥0.45. Forced NO if ring<1.40 (no close GT). GT ≤3 cm can force YES.	is_tumor_candidate, confidence, reason sentence.
ROI · confidence.py	44. Overall reconstruction confidence	One 0–1 number that blends picture quality, detection, beamformer margin, localization and focus.	overall=0.22 image_q+0.28 detection+0.15 margin+0.22 localization+0.13 focus. High≥0.75, medium≥0.45, else low. localization=e^{−d_cm/2.5} if GT exists.	ReconstructionConfidence + reasons.
Quality · tumor_taxonomy.py	45. Metadata taxonomy (labels only)	File experiments as healthy / small / medium / large, quadrant, shape. Not inferred from the image.	small if d≤2 cm, medium if d≤4 cm, else large. Quadrant from (x,y); central if √(x²+y²)<1 cm. Severity placeholder from size and BIRADS.	short_label such as small/moderate/upper-left or healthy.
Auto · discovery.py	46. Discover measurement vs metadata files	Do not treat md_list_*.mat as S-parameters.	fd_data_*.mat = cube. md_list_*/metadata_*.mat = labels. .sNp / other .mat = measurement.	List of DiscoveredFile with a kind and reason.
Auto · resolver.py	47. Pair files and pick BMID scans	A demo should cover healthy plus size/location variety, not 2000 identical clicks.	Strategies: diverse (default ~8), all, tumor_and_healthy, first. Tumor rank prefers larger d near ~3 cm offset.	ResolvedTarget list (path + scan index).
Auto · pipeline.py	48. Run the same M1→M2→M3→ROI chain unattended	Repeatable experiment. BMID: pass-through, FOV ±6 cm, 64×64. Else: savgol 7/3 + time-domain stages on.	load → preprocess → reconstruct_all → select_best_beamformer → detect_rois → refine → characterize → confidence.	Per-file evidence dict + figures.
Auto · policy.py · classifier.py	49. Threshold profiles and pass/warn/fail	Mentors can tighten or loosen gates without editing physics.	Hard fail: load error, non-finite image, energy<min, ROI area<min. Soft warn: loc error, SCR, SNR, confidence, GT-but-not-candidate. balanced: loc≤3 cm, SCR≥2, SNR≥0.5, Conf≥0.35, area≥4.	DatasetVerdict + MetricEvidence + recommended_upload.
Auto · reports.py · figures.py · cache.py · mcp_server.py	50. Write analysis pack; cache; MCP/CLI/GUI facade	One engine, three doors. Humans get markdown; machines get JSON.	Showcase = labeled tumor first, else strongest candidate. Files: latest_auto_analysis.md, latest_auto_validation.json, PNG copies.	Report paths and optional GUI handoff.

Classification — detection, size and criticality

Module / where	Step and purpose	Reason	Formulae and definitions	Expected output
Classification · features.py	51. Signal and image features for one scan	The tumour is a weak change on top of skin and fat. Features must not use the tumour's true position.	Empty-chamber subtraction. Rotational mean removed for the rsig_ copy. Breast window 0.71–5.71 ns. Image features from the blind brightest ROI.	Named feature vector. No ground-truth coordinates inside it.
Classification · scattering_math.py	52. Scattering spectrum, frequency slope, radar range	A first visit has no earlier healthy scan, so the leftover after the common-angle part is removed is described by formula.	L_i = s_i^2 / sum(s_j^2). log10 p(f) = a + b log10(f/1 GHz) on 2-6 GHz. d = R_antenna - c t_peak / 2.	15 scale-free numbers. Side check: they do not raise the official 71/100 first-scan score.
Classification · criticality.py	53. Research criticality from diameter and shape	UM-BMID has no benign/malignant label. Metadata BI-RADS is breast density.	C = (d / 2 cm)^3, times 1.5 if spiculated. Low if C <= 1, moderate if C <= 8, high above that.	low / moderate / high. Not a clinical category.
tools · train_tumor_models.py	54. Nested phantom-grouped test and the alarm cut-off	Picking the best of many tries on the test phantoms would inflate the score.	Outer 5 folds, inner 4, 3 repeats. Groups are phantom names. Cut-off maximises true-positive rate minus false-positive rate on training phantoms only.	Official first scan 71/100. Official follow-up 97/100. Saved in results/classification/final_summary.json.

Module / where	Step and purpose	Reason	Formulae and definitions	Expected output
Classification · sinogram.py · sinogram_net.py	55. Sinogram network and PCA logistic, fair-tested, not adopted	The 2022 result reads the time-by-antenna breast window with a convolution. Simulation papers reduce a radar signature with PCA, then a linear classifier.	Log magnitude in the 0.71–5.71 ns window, empty-chamber calibrated and after rotational-mean removal, resampled to 32 antennas. Two conv layers and a global average pool. PCA keeps 20 components, then logistic regression with C = 0.5.	Fair phantom-held-out ROC: network 50/100, PCA logistic 45/100. Official model remains the forest at 71/100. Numbers in results/classification/sinogram_models.json.

Required Module 3 handover (what export must contain)

Implemented in **preprocessing/handover_export.py**. The pack is: frequency_Hz and time_s; S21_raw, S21_filtered and S21_Hamming; S21_reference_subtracted and S21_clutter_removed where those stages ran; Tx_coordinates, Rx_coordinates and measurement_labels; processing_parameters and the validation report. If the file did not provide coordinates, Tx/Rx are placed on a circle of antenna_radius_m. Formats: .mat, optional .csv, .json.

If a method is not in this table, it is not part of the implemented pipeline. Classification scores are research numbers on phantoms, not a diagnosis.